

Beyond Binary

Re-imagining Age using Gender in HCI for Inclusive Design

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Abstract

As technology increasingly becomes a way access to essential services and social participation, HCI research has a fundamental responsibility to ensure digital equality and social justice across all user groups. While the field has made significant progress in understanding and incorporating gender as a social construct, leading to more inclusive design practices, the treatment of age in HCI remains largely embedded in chronological, binary concepts. Through a comparative analysis of literature in HCI on gender and ageing we identify key parallel factors in their social construction including socialisation, media representation, stereotypes, language use, and institutional frameworks. We demonstrate how these parallels can guide the evolution of age-related research and design in HCI, moving beyond simple chronological categorisations toward more nuanced, intersectional approaches. This new understanding of age has the potential to lead to more inclusive technologies that better serve users of all ages.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Social and professional topics** → **User characteristics**; • **General and reference** → **Design**.

Keywords

identity, age, gender, intersectionality, feminism

ACM Reference Format:

Alice Ashcroft and Bran Knowles. 2025. Beyond Binary: Re-imagining Age using Gender in HCI for Inclusive Design. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3706599.3719766>

1 Introduction

An enduring concern within Human Computer Interaction (HCI) research is ensuring that all technologies are inclusive, accessible, and beneficial to all users, regardless of their identity. However, the field has historically struggled with adequately addressing the complex nature of human identity, often relying on oversimplified

categorisations that fail to capture the nuanced experiences of diverse user groups and individuals. While HCI research has made significant strides in understanding and incorporating gender as a social construct, leading to more inclusive design practices, the treatment of age in HCI remains largely rooted in chronological, or binary concepts (e.g. young or old). This gap in understanding age as a complex social construct limits the effectiveness of age-related research and design in HCI. If other socially constructed identities, such as gender, can be seen as distinct from their biological alternatives (sex), and HCI can learn from this, then it seems prudent to consider whether a constructivist approach to age could be similarly productive for HCI. By examining the evolution of gender discourse in HCI and identifying key insights, we propose a number of factors which can be used for reconsidering age in HCI research and design. We argue that this approach has the potential to lead to more nuanced, inclusive, and effective technologies that better serve users across the age spectrum, ultimately contributing to more equitable and impactful technological solutions.

2 Background

HCI as a field has gradually moved away from binary gender classifications [40] (with the acknowledgement there is still more to be done [26, 50]), reflecting broader societal shifts in understanding gender identity. This transition has led to more nuanced approaches in research methodologies and design practices (e.g. in the designing of gender neutral AI voices [17] or the designing and critique of software to overcome gender bias [60]). Simultaneously, the understanding of how gender factors into technology interactions has deepened, revealing the need for a deeper understanding between gender identity, user experience, and technology design. Offenwanger et al. [40] support this with their exploration of the gender of participants within HCI, mainly CHI, research, and show evidence for the under-representation of women and non-binary participants. While their findings indicate some progress over time, they identify participant recruitment as an ongoing challenge that requires continued attention. These developments, improvements, and suggestions for further efforts in gender-related HCI research not only reflect changing societal norms but also highlight the need for more inclusive and comprehensive approaches to technology design and research.

This examination of gender representation and imbalance within HCI research demonstrates that while the field is working to improve its approach to intersectionality more widely, age remains notably absent from these discussions—other than listing chronological age when describing participants. While gender serves as

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719766>

an illustrative example of a singular diversity issue, it is essential to consider how these issues might intersect when explored alongside other identity characteristics, such as age. To date in HCI, there has been little engagement with the nuances of age itself [32]. Chronological age-based approaches in HCI often lead to overgeneralised design decisions that fail to account for individual differences [18, 32, 41, 42, 58]. For instance, when designing health monitoring applications, researchers might assume that older users invariably have poorer health, which is not always the case as age and disability are not always linked [30]. This assumption can result in interfaces that are unnecessarily simplified or even patronising to capable older users.

The field has often used 65+ as a cut-off to define “older adults”, creating a binary distinction between older and younger users. This has been critiqued by a number of HCI researchers—Durick et al. [18], Östlund [41], and Petrie [42] to name just a few—and is reflective of how gender was previously understood. This arbitrary division has implications for how technology is designed and how user experiences are understood, e.g., contributing to the problematic tendency to frame older adults through the lens of deficit [58]. Whilst there is a clear recognition within HCI of the importance of understanding ageing and designing for older adults, particularly given demographic shifts, there remain significant limitations in how the field has approached this challenge methodologically and conceptually. Much like early gender research in HCI needed to move beyond simple binary groupings to develop more nuanced, intersectional, and empowering approaches, research on ageing in HCI requires similar methodological maturation.

3 Approach

We began our inquiry into age and gender in HCI by examining how the field engages with intersectionality and gender issues. Rather than conducting a systematic review, we took an exploratory approach, starting with key HCI papers and following their references to uncover recurring themes and connections until reaching saturation, i.e. no new concepts emerged. This journey led us beyond HCI into sociological research about how age and gender identities are formed. To make sense of these findings, and through a process of iteration, we created a comparative table (Table 1) that shows how both age and gender identities are shaped by conceptually similar identity factors.

We stress that this work is preliminary, offering a glimpse into the potential for beneficial synergy between previously siloed sub-areas of HCI research. We encourage more systematic analysis of the literature going forward, with a view to careful examination of the similarities and differences between the constructs of age and gender, as well as the implications of their intersection with each other and other identity characteristics. While this paper draws from gender as an illustrative example of social construction in HCI, we see important work in expanding this analysis to integrate concepts relating to the social construction of, e.g., race and disability.

4 Exploring Identity

In what follows, we discuss how each of the seven identity factors identified in the literature may impact HCI. Our aim is to illuminate

advantages the HCI and Ageing community may have to gain from viewing age in the same way that gender has come to be viewed.

4.1 Biological Factors

While gender and biological sex are distinct concepts [53], they are often conflated in societal understanding. The majority of the population identifies as cisgender, meaning their gender identity aligns with their biological sex. However, it is crucial to acknowledge that while gender and sex can be linked for many individuals, this is not universally true. Similarly, there is an important biological component to ageing, with physiological changes occurring over time [10]; yet individuals experience these changes at different rates due to a variety of genetic and environmental factors, so the process of ageing can look radically different for two individuals of the same chronological age.

While biological factors no doubt influence technology interactions (e.g. phones are typically bigger than ideal for female hands [16]), it is dangerous to assume differences between user groups are biologically determined. Historically, sexist and ageist rhetoric emphasises biological difference while de-emphasising the ways that gender and age are socially constructed. We explore these constructions in our discussion of the remaining identity factors.

4.2 Socialisation and Societal Norms

Through the process of socialisation, individuals internalise societal norms and expectations. For gender, this involves learning behaviours and expressions deemed appropriate for one’s assigned sex [21, 37, 38]. Likewise, age identity is shaped by cultural and societal norms that dictate age-appropriate behaviours and roles [46]. These norms vary across cultures and historical periods, influencing how individuals perceive and express their age identity [14].

Societal expectations of age and gender significantly influence technology interactions in HCI. Males, for example, demonstrate high task completion rates and more positive ratings of features [44], though it is likely that such findings relate to life-long gender differences in a range of computer experiences and internalised expectations around male technology self-efficacy [56]. Similarly, while old age is significant predictor of digital exclusion [24], there is evidence that older adults judge one another according to internalised beliefs about age-appropriate technology use [52]. These findings emphasise the need for HCI practitioners to critically examine socialised norms when designing technologies, i.e. how a person’s technology interactions reflect internalisation of beliefs about what one *should* do.

4.3 Stereotypes

Stereotypes often serve the purpose of perpetuating norms. For gender, social circles reinforce or challenge societal expectations through interactions, role modelling, and explicit expectations [5, 27, 37]. In the context of age identity, perceptions of ageing shaped by societal attitudes and stereotypes influence how individuals view themselves and others as they grow older [4, 35, 62, 64]. Building upon the influence of family and peer groups on gender and age identity formation, we can draw parallels between the works of Vauclair et al. [55], Kornienko et al. [34], and Loy and Norland [39]. Vauclair et al. [55]’s research on older people’s perceptions

Identity Factors	How Gender Is Constructed	How Age Is Constructed
Biological Factors	Gender and sex can <i>often</i> be linked (i.e. the majority of the population are cis-gendered), but they are not always [53].	Ageing involves physiological changes in cells and tissues which can, over a long enough period of time, lead to biological deterioration. Rates of change are affected by genetic predisposition [25] as well as environmental conditions [8], so there is significant heterogeneity in biological ageing relative to chronological age.
Socialisation	Through the process of socialisation, individuals learn and internalise societal norms and expectations related to their assigned sex. This includes how to express emotions, interests, and behaviours deemed appropriate for their gender [21, 37, 38].	Notions of age are construed according to cultural and societal norms which dictate age-appropriate behaviours, expectations, and roles. These norms vary across cultures and historical periods [46].
Stereotypes	Families and peer groups play a significant role in gender construction, as they reinforce or challenge societal gender expectations through social interactions, role modelling, and expectations [5, 27, 37].	Age is constructed through perceptions of ageing, with societal attitudes and stereotypes shaping how individuals view themselves and others as they grow older [4, 35, 62, 64].
Media Representation	Media portrayals often perpetuate stereotypes and reinforce gender norms. Representation in media influences how people perceive and identify with gender roles and expressions [22, 61].	Individuals may identify with the experiences and values of their generation, contributing to a sense of a shared identity which is perpetuated by the media [7].
Language	Gender is constructed through language, with specific words, phrases, and linguistic conventions associated with particular genders. These linguistic choices can shape and reinforce gender stereotypes and roles [28, 51].	Language relating to age categories can subtly reinforce ageist stereotypes. Individuals engage particular linguistic strategies in affiliating with or distancing from age categories and their associated stereotypes [29].
Life Course Events	Life events such as income, opportunities, coming out as trans, etc., are all affected by gender and change throughout one's life [9, 45, 46].	The age at which individuals are expected to experience major life course events, such as marriage, parenthood, retirement, and other transitions, affect how people see themselves in relation to their roles within society [9, 45, 46].
Institutional and Legal Norms	Gender construction is influenced by institutional and legal norms, such as marriage laws, workplace policies, and educational systems [13]. These norms can either perpetuate or challenge traditional gender roles. Gender is one of the protected characteristics under the Equality Act.	Age discrimination legislation in the UK, such as the Equality Act 2010, is an identity factor related to older adults. Age discrimination laws protect the rights and dignity of older adults, impacting their self-identity and societal roles as they age [31, 48].

Table 1: A Comparison of Literature between How Gender and Age are Socially Constructed.

of age discrimination highlights the importance of both personal meta-perceptions and societal meta-perceptions in shaping age identity. This approach resonates with Kornienko et al. [34]'s study on peer influences on gender identity in adolescents which found significant peer influence on between-gender components of gender identity. Both studies emphasise the role of social context in identity formation, whether it's age or gender.

In the context of HCI, peer influences and perceptions of ageing significantly impact technology adoption among older adults. Feist et al. [23]'s survey showed that older adults' technology comfort levels vary based on their social connections, while Shi et al. [49] demonstrates how peer influence affects technology adoption rates. Building on Renaud and Van Biljon [43]'s work on technology acceptance, peer experiences can enhance or diminish perceived usefulness and ease of use. Interestingly, HCI research shows that while some older adults internalise negative negative stereotypes

around digital proficiency of people their age [63] and justify their own disengagement through reference to these stereotypes [33], others actively distance themselves from these stereotypes [36] and leverage their digital proficiency to reject being labelled as 'old'. As noted by Bardzell [6] of gender and by Vines et al. [58] of age, it is important that HCI reflect on whether/how research and design inadvertently perpetuates negative stereotypes, and how in so doing it reinforces social inequities.

4.4 Media Representation and Grouping Identity

Media portrayals significantly influence the construction of both gender and age identities. For gender, media representations often perpetuate stereotypes and reinforce gender norms, shaping how people perceive and identify with gender roles and expressions [61]. The documentary "Disclosure" [22] provides a compelling

examination of these media influences on gender identity. But in the context of age, media representations contribute to the formation of generational identities, where individuals identify with the experiences and values associated with their generation [7]. These shared generational experiences, often perpetuated and reinforced by media portrayals, create a collective identity that influences how individuals perceive themselves and others within their age cohort.

Media representations and generational identity distinctly shape how age groups interact with technology. Vittadini et al. [59] shows how early media experiences shape technological practices throughout life, suggesting that youth exposure to technology influences later adoption patterns and comfort levels. Furthermore, Williams et al. [66]’s study on young people’s responses to media constructions of “Generation X” provides insight into how generational labels and stereotypes can affect self-perception and technology use. In HCI, this could relate to how different age groups approach new technologies based on their perceived generational identity. For example, Petrie [42]’s development of personas of older adults based on well-known celebrities aims to challenge these generational stereotypes in technology design. This innovative approach to personas demonstrates how HCI researchers can actively work to disrupt age-based stereotypes, suggesting that careful consideration of media representation and its influence can lead to more nuanced and inclusive design practices.

4.5 Language

Language plays a crucial role in constructing both gender and age identities. In gender studies, researchers have identified specific words, phrases, and linguistic conventions associated with varying gender expressions, which can shape and reinforce gender stereotypes and roles [28, 51]. Similarly, in age studies, Jones [29] explores how older people use language to distance themselves from the category of “older people,” employing critical discursive psychology [19, 20, 65] to analyse these linguistic strategies. This research highlights the importance of self-reflection and “rejecting normative assumptions about being older” [29]. The linguistic construction of both gender and age identities demonstrates the power of language in shaping how individuals perceive and present themselves within society [15, 47].

In the context of HCI, the language used in technology interfaces and interactions can significantly impact how older adults perceive themselves and their relationship with technology. For example, Renaud and Van Biljon [43]’s work on technology acceptance among older adults highlights how the language used in user interfaces can influence perceived usefulness and ease of use. When technology employs language that assumes deficits or lack of experience among older users, it may reinforce negative self-perceptions and discourage engagement. Furthermore, Feist et al. [23]’s survey findings suggest that when technology is presented using language that emphasises connection and community, older adults may feel more positively about their ability to engage with it. This aligns with Jones [29]’s observations on how older people linguistically distance themselves from ageist categories, suggesting that HCI designs that avoid patronising or overly simplistic language may better support older adults’ positive self-perception and technology adoption.

4.6 Life Course Events

Expectations regarding gender will shift depending on when life course events happen for each individual, e.g. “expected” ages to get married (which may differ for men and women) [14]. Gender and age are intrinsically linked in this way. Furthermore, there are “gendered” life course events that will affect individuals such as coming out as trans, the rights to parental leave differing by gender and country, and so many more. Each of these needs to be further understood to uncover how this will impact HCI. Major life transitions such as marriage, parenthood, retirement, and other milestones affect how individuals perceive themselves and their roles within society [9, 45, 46]. These events often mark shifts in social status and responsibilities, contributing to the dynamic nature of age identity throughout the lifespan. The impact of life course events on age identity highlights the temporal and contextual aspects of identity formation, which can also be observed in the evolving nature of gender identity across different life stages.

Life course events can significantly shape interactions with technology, for example based on when and how they encounter different innovations [42]. For instance, exposure to workplace technologies may be interrupted or altered by events like maternity leave or early retirement—potentially creating lasting gaps in technological familiarity. Similarly, major life transitions like moving into assisted living facilities might introduce individuals to specific healthcare technologies at a time when learning new systems feels particularly challenging. These intersections between life events and technological evolution can create distinct patterns of technology adoption and comfort that persist far beyond the initial event itself.

4.7 Institutional and Legal Norms

Both gender and age identities are influenced by institutional and legal frameworks. For gender, construction is shaped by marriage laws, workplace policies, and educational systems [13], which can either perpetuate or challenge traditional gender roles. Both characteristics are protected by the UK equality law which protect the rights and dignity of older adults, impacting their self-identity and societal roles as they age [31, 48], and protect people from gender discrimination. Although, of course, discrimination still takes place, the laws are there as an attempt to stop this.

When it comes to HCI, institutional and legal norms significantly impact the design and implementation of technologies for older adults, e.g. AI recruitment tools which must incorporate fairness measures to prevent age-based discrimination. Legislation can also be understood *alongside* professional responsibility, whether within a body or on a personal level, to understand how legal protections for older adults should be understood. For example, UNHCR [54] highlights the need to ensure digital accessibility and inclusive design for older persons, which aligns with both legal requirements and professional computing ethics. Professional bodies in computing and HCI, such as the ACM [1] and IEEE [2], have established ethical guidelines that specifically address responsibilities toward marginalised groups, including older adults. These professional standards complement legal frameworks by promoting inclusive design practices, accessibility requirements, and ethical considerations in technology development. This combined approach of legal compliance and professional ethics creates a more comprehensive

framework for ensuring technology serves and protects older adults' rights and dignity in the digital age.

5 Discussion

Our review of the literature highlighted the need to move beyond simplistic categorisations of both age and gender, acknowledging the complex nature of these identity characteristics and how they are different to their biological counterparts. An exploration of potential implications for design and addressing identity related challenges is explored in this section.

5.1 An Intersectional Lens on Age in HCI

Intersectionality, built upon Black Feminist Theories [3, 57], is a crucial concept in understanding both gender and age identities, both biologically and socially. In gender studies, Intersectionality recognises that gender construction is influenced by other aspects of identity, including race, ethnicity, class, and sexuality [6, 57]. The intersection of these identities leads to unique experiences and constructions of gender through lived experience, e.g., with some experiencing multiple marginalisation.

Age is another important variable that comprises a person's intersectional identity [12], though HCI research on ageing tends to explore age factors in isolation from other identity factors. In contrast, an Intersectional approach would more deeply examine how an individual's experience of age and ageing is shaped by membership in *multiple* social categories and by the structural constraints each category encounters. By attending to these intersections, HCI researchers and designers can move beyond one-dimensional understandings of age or gender, to recognise that all people are multidimensional—at times experiencing particular aspects of identity more saliently than others [11], but always experiencing their lives from a particular perspective informed by their multiple identities.

5.2 Implications for Design and Development

Understanding the implications of identity-related research in HCI for design and development is important for several reasons, namely for designing technology that resonates with a broad user base, creating equal opportunity to engage in non-essential and life enhancing digital services, and ensuring that technological advancements serve to reduce, rather than reinforce or exacerbate, existing social inequalities. Collaboration between researchers and practitioners might include investigating how an integrated framework aligns with existing design methodologies and whether it necessitates new approaches. For example, does the chosen methodology allow for participants to embody and express the perspectives of their multiple identities, and for researchers to consider how different combinations of identity factors shape technology interaction, engagement, satisfaction, and/or adoption? It is our aim that the age identity factors explored within this paper are used as prompts for reflexive practices in all HCI research.

Future research is needed to delve into the practical applications of an integrated framework in the design and development processes. This involves exploring how considerations of age, gender, and other identity factors can inform user interface design, feature development, and overall user experience. We anticipate particular

progress could be made through deeper engagement with more diverse older adult constituencies in HCI research on ageing, as the process of ageing allows for differentially experienced structural constraints encountered by people of different social categories to cumulate to significance [32]. By understanding how a person's intersectional and multidimensional experience of age informs their late life technology relations, we may begin to see patterns in how age and gender (and other aspects of identity) can be linked in practical terms to the design of more equitable technologies.

6 Conclusion

This paper has provided an exploration of age and gender considerations in the context of HCI. The complex nature of age and gender identity construction has been examined, drawing parallels between the two and emphasising the importance of an Intersectional approach. By reviewing existing literature on age and gender in HCI, this paper has laid the groundwork for understanding the complexities of identity within HCI, and presenting recommendations based upon this. Just as the understanding of gender in HCI has evolved to embrace its socially constructed factors, recognising age as similarly constructed opens new doors for more nuanced and equitable technology design. This perspective challenges HCI researchers and designers to create technologies that respect and respond to the full complexity of human identity rather than reducing users to simplified age-based stereotypes, and allows the field to explore interesting and complicated questions about what different people may bring to their interactions with technology.

Acknowledgments

This work is supported by the EPSRC funded grant Equity for the Older: Beyond Digital Access (EP/W025337/1).

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